

Quadratic Cohomology

G finite abstract simplicial complex.

$$\chi_2(G) = \sum_{x,y \in G} \omega(x)\omega(y)$$

quadratic characteristic Wu

$$df(x,y) = d_x f(x,y) + \omega(x)d_y(f,y)$$

exterior simplicial derivative on
 $\Lambda = \{f: \{(x,y) \in G \times G, x \cap y \in G\} \rightarrow \mathbb{R}\}$

$$D = d + d^*$$

Dirac
 $H = D^2 = dd^* + d^*d$ Hodge
 $b_k = \dim(\ker(H_k))$ Betti

We get analog stories to Euler case:

Euler-Poincare

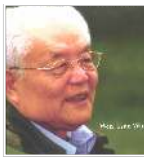
Gauss-Bonnet

Brouwer-Lefschetz

Poincaré-Hopf



Leonhard Euler
1707-1783



Wenjun Wu
1919-2017

x	x	$\omega(x)$
●	{1}	1
●	{2}	1
—	{1,2}	-1

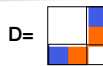


$$b = (1,0)$$

Betti vector

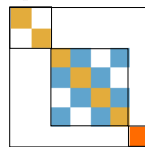
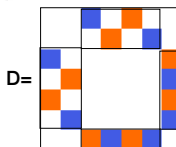
$$\chi(G) = 1$$

constant 0-forms are harmonic



(x,y)	x	y	$\omega(x)\omega(y)$
● ●	{1}	{1}	1
● ●	{2}	{2}	1
● —	{1}	{1,2}	-1
● —	{2}	{1,2}	-1
— ●	{1,2}	{1}	-1
— ●	{1,2}	{2}	-1
— —	{1,2}	{1,2}	1

$$\chi_2(G) = -1$$



$$b = (0,1,0)$$

quadratic Betti vector

constant 1-forms are harmonic

$G = \{\{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\}$

the only harmonic quadratic differential forms are 2-forms

linear

$$\chi(G) = 1$$



$$D =$$

$$H =$$

Λ is 7-dimensional
 $b = (1,0,0)$

quadratic

$$\chi_2(G) = 1$$



$$D =$$

$$H =$$

Λ is 37-dimensional
 $b = (0,0,1,0,0)$

Valuations



Hadwiger
1908-1981



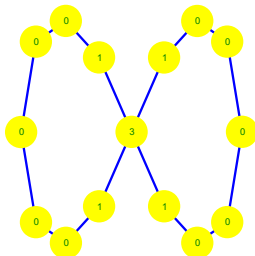
Dan Klein

J-C. Rota

Basis for linear valuations : f_k
 Valuations are $q+1$ dimensional

Basis for quadratic valuations : f_{kl}
 Valuations are $(q+1)(q+2)/2$ dimensional

Any valuation comes with a Gauss-Bonnet theorem etc.



Gauss-Bonnet

$$K(v) = \sum_{x \sim y, v \in x} \frac{\omega(x)\omega(y)}{|x|}$$

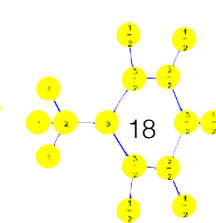
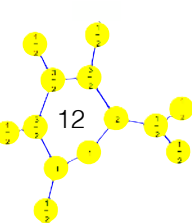
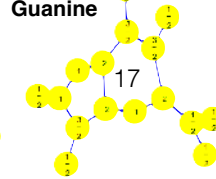
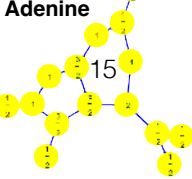
curvature

$$\sum_{v \in V} K(v) = \chi_2(G)$$

Gauss-Bonnet for quadratic characteristic

Adenine

Guanine



Cytosine

Thymine

Topological

Quadratic cohomology is an example of a topological invariant, not homotopy invariant!

Bott Introduced term "Topological Invariant"

TWO NEW COMBINATORIAL INVARIANTS FOR POLYTOPES
 by DANIEL GALE

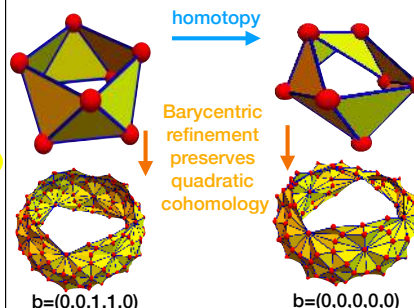


1923-2005

Example:

Cylinder
 $b = (0,0,1,1,0)$

Möbius strip
 $b = (0,0,0,0,0)$



Boundary Formula

Let G be a q-manifold with boundary

$$\chi_2(G) = \chi(G) - \chi(\delta G)$$

$$= \chi(\text{int}(G))$$

Proof: a manifold with boundary is characterized by having every $U(x)$ for x in the interior to have a sphere boundary $S(x)$ so that

$\chi_2(U(x)) = \chi_2(B(x)) - \chi_2(S(x)) = \omega(x)$
 which agrees with $\chi(U(x))$. Therefore, using inclusion exclusion, the linear and quadratic characteristic agree on open sets.

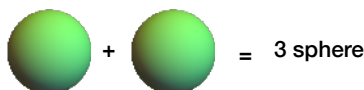
q even

$$\chi_2(G) = \chi(G)$$

q odd

$$\chi_2(G) = -\frac{\chi(\delta G)}{2}$$

(glue 2 disjoint copies of G along δG)



Cubic Cohomology

G finite abstract simplicial complex.

$$\chi_3(G) = \sum_{x,y,z \in G} \omega(x)\omega(y)\omega(z)$$

cubic characteristic

$$df(x,y,z) = d_x f(x,y,z) + \omega(x)d_y(f,y,z) + \omega(x)\omega(y)d_z(f,y,z)$$

exterior simplicial derivative on

$$D = d + d^*$$

$$H = D^2 = dd^* + d^*d$$

$$b_k = \dim(\ker(H_k))$$

Dirac
 Hodge
 Betti

Let G be a q-manifold with boundary

$$\chi_3(G) = \chi(G) + \chi(\delta G)$$

All balls have Cubic characteristic 1

q even

$$\chi_3(G) = \chi(G)$$

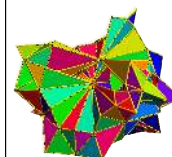
q odd

$$\chi_3(G) = \frac{\chi(\delta G)}{2}$$

Complete Complexes

Complex	Euler	Betti	Wu	Betti	Cubic	Betti
K1	1	(1)	1	(1)	1	(1)
K2	1	(1,0)	-1	(0,1,0)	1	(0,0,1,0)
K3	1	(1,0,0)	1	(0,0,1,0,0)	1	(0,0,0,1,0,0)

Dunce Hat



homotopic to 1
 not contractible



$$\chi_1(G) = 1 \quad b_1(G) = (1,0,0)$$

$$\chi_2(G) = 1 \quad b_2(G) = (0,0,0,1,2)$$

$$\chi_3(G) = 1 \quad b_3(G) = (0,0,0,0,0,1,2)$$

higher cohomology sees non-contractibility